



QuntrolSphere

Pvt. Ltd.

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QKD Network Primer

The QKD networks primer introduces the operational dynamics of Quantum Key Distribution (QKD) implementation. The primer is designed to provide users with hands-on experience in building and interacting with optical systems, thereby bridging the gap between theoretical knowledge and practical application. The primer program also aligns with AICTE and the National Quantum Mission's initiatives for capacity building and also provides a comprehensive solution to establish Centres of Excellence or Quantum Labs, in academic and R&D institutes.

Supported Experiments & Features

- **Source and Channel Analysis**
 - Optical source characterization
 - Controlled channel noise addition:
 - Experiments with variable optical loss
 - Degraded fiber conditions
 - Additional optical source at the same wavelength (λ)
- **Tx-Rx Pattern Analysis**
 - Generation and observation of multiple transmission patterns
 - Correlation of transmitted and GATE patterns
- **QKD implementation leading to key generation**
 - Photon arrival histogram generation and analysis
 - Impact of loss, noise, and timing offsets on QKD implementation
 - Study of histogram with variation in key parameters (detector efficiency, channel attenuation etc.)



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- **Security Attack Emulation**

- Use of additional optical Tx/Rx modules to emulate adversarial behavior
- Demonstration of blinding attacks
- Beam splitter (splitting) attack implementation

- **Instrumentation**

- Combinations of breadboard with PIN photodetectors with and without TIA.
- Design of TIA circuitry with different gain.
- Delay line implementation.

- **QKD network digital twin**

- Hands-on towards classical-QKD network integration, key access mechanism, QKD applications and use cases development, networking control and monitoring etc.
- Create a virtual network of QKD nodes with different topologies (e.g., linear, mesh, star) and simulate the distribution of quantum keys across the network.
- Mini project: develop a QKD application

Learning outcomes :

- Fundamentals of elementary blocks of a fibre based QKD system.
- QKD experiments under different test conditions, including attacks.
- Integration of QKD with the classical network.
- QKD application development and key access mechanism



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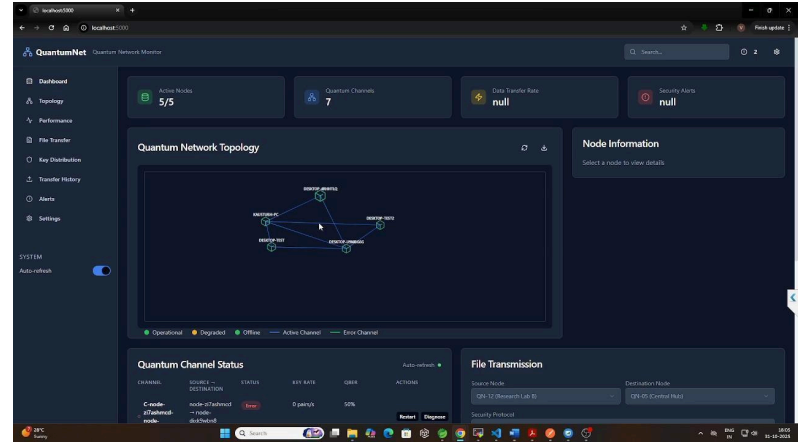


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QKD Networks Primer



QKD network digital twin

Support :

- Course-aligned academic support to ensure smooth integration into existing programs.
 - Structured experiments
 - Alignment with UG / PG / research curriculum
- Support in Setting up lab

Delivery time

Hardware configurations allow **two students to share one unit**, enabling efficient resource utilization. *(Final pricing is subject to customization and institutional requirements.)*

- **Ready for delivery:** April 2026 onwards (negotiable based on order size)
- **Lead time:** 1 Month